Stefano Blando

stefano.blando@santannapisa.it — LinkedIn — GitHub — Website — Scholar

Profile — PhD student in Artificial Intelligence within the National PhD Program (Scuola Superiore Sant'Anna & University of Pisa). Research at the intersection of AI, agent-based modeling, and economics: adaptive multi-agent systems, differentiable agent-based models, LLM-based generative agents, and robust quantitative methods for financial, socio-economic, and textual data.

Education

PhD in Artificial Intelligence for Society *Nov 2025 – Present*
National PhD Program – Scuola Superiore Sant'Anna & University of Pisa
Supervisors: Prof. Andrea Vandin, Prof. Giorgio Fagiolo, Prof. Daniele Giachini
Research focus: "Learning How to Learn: Adaptive Cognitive Architectures for Economic Network Formation" — differentiable agent-based models, LLM-based generative agents, multi-agent learning, statistical model checking, network dynamics, and robust statistics.

Second-Level Master in Customer Experience, Statistics, ML and AI *Feb 2025 – Feb 2026*
University of Rome Tor Vergata + SAS Academy
Final Grade: 110/110 cum laude
Thesis: "Network Topology Analysis and Machine Learning Techniques for Systemic Risk Prediction in U.S. Equity Markets"
Advanced program in data science, machine learning, and artificial intelligence with SAS Academic Specialization in Advanced Data Analytics and Machine Learning Engineering.

MSc in Financial Markets and Financial Intermediaries *Oct 2022 – Apr 2025*
University of Rome Tor Vergata, School of Economics and Finance
Final Grade: 108/110
Thesis: "High-dimensional Robust Portfolio Optimization Under Contamination: A Factor-Analytic Approach"
Supervisor: Prof. Alessio Farcomeni

BSc in Governance and International Relations *Sep 2019 – Sep 2022*
University of Rome Tor Vergata, School of Law
Final Grade: 110/110 cum laude
Thesis: "Consumer Choice Under Uncertainty: From Homo Oeconomicus to Homo Temperatus"
Supervisor: Prof. Gustavo Piga
Erasmus+ Exchange: University of Paris Est-Créteil, France *Aug 2021 – Feb 2022*
Selected courses: Trade Data Analysis, Sustainable Development, Business and AI, International Strategy

BA in Philosophy (not completed) *Sep 2015 – Sep 2017*
La Sapienza University of Rome
Coursework in logic, epistemology, and analytical philosophy. Program paused for personal reasons.

Publications & Submissions

Peer-reviewed

- S. Blando, G. Fagiolo, D. Giachini, A. Vandin, E. Ivanaj. **"Statistical model checking of the Island Model: an established economic agent-based model of endogenous growth"**. *MARS @ ETAPS 2026*, Turin, Italy (EPTCS proceedings).
- S. Blando, D. F. Iezzi. **"A Multi-Method Validation Framework for Large-Scale Multilingual Text Analytics"**. *JADT 2026*, Palermo, Italy (Proceedings ISBN 978-88-5509-883-0). **VADISTAT Award 2026**.

Under review

- S. Blando, A. Farcomeni. **"Robust Portfolio Optimization Under Systematic Market Disruptions: A Factor-Analytic Approach"**. Submitted to *Computational Economics* (Mar 2026).
- S. Blando, G. Fagiolo, M. Napoletano, T. Treibich, A. Vandin. **"Statistical Model Checking of the Keynes+Schumpeter Model: A Transient Sensitivity Analysis of a Macroeconomic ABM"**. arXiv:2605.10447; journal version in preparation.
- S. Blando, E. Guerrazzi, R. Porcedda, G. Squillace, M. Tschaikowski, A. Vandin. **"Towards Agentic Agent-based Models: Feasibility, Performance, and Statistical Model Checking"**. Submitted to *AISoLA 2026*; arXiv:2607.17948.

Grants & Awards

- **VADISTAT Award for Young Researchers (2026):** "Associazione VADISTAT — Per Simona Balbi" award for the best scientific contribution by researchers under 35, 18th International Conference JADT 2026, Palermo.
- **CINECA ISCRA C Grant (2026):** awarded compute resources on the Italian national supercomputing infrastructure (Leonardo) for a project on differentiable, tensor-native simulative models.
- **University of Pisa Doctoral Grant for Student-Led Scientific Events (2026):** funding to co-organize COMPASS, an interdisciplinary scientific workshop on complexity, markets, policy, and AI in social systems.
- **2nd Place, AI Data Hackathon 2024 (Hera Group):** predictive model for gas leak detection using CTGAN data augmentation and SHAP explainability.
- **Finalist, Alpine Climate Data Challenge 2025:** selected among 150+ teams for climate forecasting with Copernicus ECMWF, NOAA, and Arpa Piemonte datasets.
- **1st Place, Huawei DigitALL Night 2024:** competition on digital technology expertise.
- **Selected, Bertelsmann Next Gen Tech Booster:** chosen from 17,000+ candidates for the Generative AI nanodegree with Udacity.
- **Tutorship Grants (multiple editions):** University of Rome Tor Vergata — competitive merit-based tutorship grants awarded for MSc course teaching assistance, Data Processing Center (CED) IT support, and library services.

Academic Service

Peer Review

- Reviewer, *Journal of Artificial Societies and Social Simulation* (JASSS) — 2026
- Reviewer, *Annual Modeling and Simulation Conference* (ANNSIM) — 2026
- Subreviewer, *QEST + FORMATS* — 2026

Event Organization

- **Co-organizer and panel speaker, COMPASS — Complexity, Markets, Policy, and AI in Social Systems.** Scuola Superiore Sant'Anna, Pisa, 18 May 2026. Interdisciplinary workshop organized within the National PhD Program in AI: three keynote lectures, three thematic panels, and a multi-institutional network spanning UniPi, SSSA, CNR, UCL, and Penn State. Funded by a University of Pisa doctoral grant.

Talks & Presentations

- **MARS @ ETAPS 2026**, Turin, Italy — April 2026. Paper presentation: statistical model checking of the Island Model.
- **JADT 2026**, Palermo, Italy — July 2026. Paper presentation: multi-method validation framework for multilingual text analytics (VADISTAT Award).
- **COMPASS 2026**, Scuola Superiore Sant'Anna, Pisa — May 2026. Panel speaker.

Academic Training & Schools

- **SoBigData Summer School 2026 — "Data Analysis and AI in Sport, Health and Wellbeing"** — Baratti (Piombino), Italy, 21–27 June 2026
  Interdisciplinary training in data analysis, machine learning, and AI for societal challenges, with visual explainable AI and a team challenge on sports analytics
- **Workshop in Economic Complexity** — Enrico Fermi Research Center, Rome, July 2025
  Advanced methodologies in complexity economics and network analysis
- **Oxford Summer School in Economic Networks (OSSEN)** — University of Oxford, June 2025
  Lectures and coding tutorials in network formation theory and complex systems, with Doyne Farmer, Rama Cont, Aaron Clauset, Renaud Lambiotte
- **YERUN Summer School in AI-Empowered Management Consulting** — Blended Intensive Program (Erasmus+)
  Collaborative project with students from Spain, Germany, UK, and Norway on AI applications in strategic business diversification
- **PhD coursework:** Computational Economics and Agent-Based Macroeconomics (Fagiolo, Roventini, Vandin), Statistical Learning for Large Datasets (Chiaromonte), Networks and Complex Systems (Squartini, Garlaschelli)
- **Erasmus+ Exchange:** University of Paris Est-Créteil, France — 6-month program on Business Intelligence, AI, and International Strategy

Teaching & Academic Support

Academic Tutor, School of Economics and Finance (Tor Vergata) *Oct 2022 – Jan 2025*

- **MSc Teaching Assistant (Prof. Andrea Appolloni):** Supported the master's course *Sustainable Supply Chain and Operations Management*, assisting with case study discussions, exam organization, and student team projects.
- **Data Processing Center (CED) IT Support:** Provided technical and IT assistance to students and faculty, including workstation maintenance, software setup, and lab support.
- **Student Academic Advisory & Info Desk:** Managed the student orientation desk, providing guidance on study plans, degree requirements, and academic procedures.

Library & Academic Tutor, School of Law (Tor Vergata) *Mar 2020 – Sep 2022*

- Provided front-desk assistance, bibliographic reference services, and study-room coordination for faculty students and patrons.

Selected Applied & Software Projects

RiskSentinel — Microsoft AI Dev Days Hackathon 2026 *Feb 2026*

- Multi-agent systemic-risk simulator on S&P 500 networks (210 stocks, 3,081 daily snapshots) with contagion models including DebtRank, Linear Threshold, and Cascade Removal.
- Interactive visualization and reporting built with Streamlit, Plotly, and the Microsoft Agent Framework on Azure OpenAI; tested core engine with 41 passing unit tests.

Multi-Agent Orchestration *Mar 2026*

- Event-driven multi-agent orchestration system with phase-specific strategies, structured action interfaces, and runtime state tracking for real-time decision making.
- Persistence, metrics, and replay analysis for execution reliability under noisy and time-constrained environments.

Advanced Recommender System *Jun 2025*

- Listwise ranking recommendation system with LightGBM/XGBoost and NDCG@K optimization.
- GroupKFold validation and ensemble weighting for ranking performance.

Real Estate AI Agent *Apr 2025*

- Autonomous AI agent for property-market analysis combining predictive valuation models with LLM-based orchestration for task-level reasoning.
- Data acquisition and trend-analysis workflows for real-estate pricing insights and interactive natural-language querying.

Gas Network Risk Forecasting — Hera Group Hackathon (2nd Place) *Nov 2024*

- Hybrid ML pipeline for gas leak prediction with geospatial-temporal features.
- Synthetic data generation (CTGAN, TimeGAN) and SHAP-based interpretability to improve minority-class detection.

Leadership & Extracurricular Activities

Secretary General, Starting Finance Club Tor Vergata *2023 – 2024*

- Coordinated a team of 70+ members and organized 10+ events including workshops, talks, and presentations with financial sector professionals
- Led macroeconomic and sectoral analysis initiatives
- Facilitated collaboration with Deltahedge for financial analysis and outlook development; coordinated a visit to Google EMEA HQ Dublin

Mentoring & Community Engagement

- **Mentors4U Program:** Selected as mentee in Italy's premier non-profit mentoring program for high-potential university students
- **Academic & Peer Mentoring:** Mentored junior students in quantitative methods, data analysis, and university orientation within student-led initiatives

Professional Development & Certifications

AI, Machine and Deep Learning: Gen AI nanodegree (Udacity), Applied Data Science Lab (WQU), Deep Learning Lab (WQU), Deep Learning with PyTorch (365DS), Deep Learning with TensorFlow 2 (365DS), Machine Learning (MATLAB), Deep Learning (MATLAB).
SAS Certifications: AI Foundation Knowledge, Optimization for Data Science and AI, SAS Base Programming, Machine Learning using SAS Viya, Interactive Machine Learning, SAS Viya, Text Analytics, Programming 1 & 2
Advanced Quantitative Finance: ARPM Quant Bootcamp, Bloomberg Market Concepts & ESG & Finance, Financial Engineering Masterclass (Starting Finance)
Business & Strategy: McKinsey Forward Program, Candriam Academy (ESG, Circular Economy, Climate Change)
Technical Skills: 365 Data Science (Python, SQL, Advanced Excel, Statistics), MathWorks ML & DL Onramp
Specialized Training: ECB Monetary Policy (20 hrs with Lead Economist), Time Series Symposium Villa Mondragone, SACE Credit Risk Analysis, Sustainable Finance and Investments Workshop, Quantitative Finance at Work (2 editions), Storytelling Workshop, Bloomberg Workshops (Discover Bloomberg, Bloomberg Horizon)

Technical Skills

- **Languages & Scientific Computing:** Python, JAX, PyTorch, R, MATLAB, SQL, SAS, LaTeX, Git
- **Machine Learning & AI:** differentiable programming, graph neural networks, reinforcement learning, simulation-based inference, multimodal AI, explainable AI, LLM fine-tuning and RAG, multi-agent orchestration
- **Statistical Methods:** robust estimation, high-dimensional inference, factor models, time-series analysis, functional data analysis, sufficient dimension reduction, universal inference, causal inference, Monte Carlo methods
- **NLP & Text Analytics:** text mining, semantic network analysis, topic modeling, sentiment analysis, transformer-based classification
- **Complex Systems & Networks:** multilayer networks, network topology, community detection, contagion and herding dynamics, temporal networks, financial network modeling
- **Agent-Based Simulation:** differentiable simulation, multi-agent systems, statistical model checking, emergent behavior analysis, counterfactual analysis
- **Quantitative Finance & Risk:** systemic risk modeling, portfolio optimization, stress testing, financial econometrics, scenario analysis, model validation
- **Infrastructure & Platforms:** SLURM/HPC clusters (CINECA Leonardo), GPU computing, distributed computing, MLOps/LLMOps, Bloomberg Terminal, LSEG Workspace

Languages

Italian (Native), English (Fluent, C1/C2), French (B1), Spanish (A2), German (A1)